

Arushi Jha

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MS Data Science graduate with 2+ years of experience in predictive modeling, feature engineering, and machine learning across financial and enterprise domains. Skilled in Python, SQL, and statistical analysis with a track record of delivering data-driven insights that improve decision-making, reduce operational risk, and enable measurable business outcomes.

Education

University of Colorado Boulder, USA | MS in Data Science | GPA: 3.9/4.0 Aug 2024 – May 2026

Relevant Coursework: Machine Learning, NLP, Statistical Modeling, Data Mining, Datacenter Scale Computing

VIT, India | Bachelor of Technology in Computer Science | GPA: 3.7/4.0 Aug 2018 – May 2022

Technical Skills

Programming: Python, SQL, R, Java

Machine Learning: Scikit-learn, XGBoost, TensorFlow, PyTorch, SMOTE, Classification, Regression, Clustering, PCA, Association Rule Mining

GenAI & AI: LLMs, RAG, Agentic AI, Multi-Agent Systems, LangChain, LangGraph, OpenAI GPT-4o, Embeddings, Semantic Search, FAISS, Pinecone, Sentence-Transformers, Vector Databases, TF-IDF

Feature Engineering & Analysis: Pandas, NumPy, Statistical Analysis, Predictive Modeling, Model Evaluation, Feature Engineering, NLP

Visualization & BI: Tableau, Power BI, Streamlit, Matplotlib, Seaborn, Excel

Data Engineering & Cloud: PostgreSQL, Snowflake, Databricks, Azure, GCP, Spark, Redis, Docker, FastAPI, ETL Pipelines

Tools & Platforms: Git, LangSmith, RAGAs, Jenkins, SonarQube, Jira, Pytest

Experience

Data Scientist Intern, Parlay Finance Jan 2026 – Apr 2026

- Designed a weighted scoring engine encoding **8** SBA domain parameters across **341** production loan inquiries, generating ground-truth labels that enabled downstream ML training with **99%** F1 score.
- Applied SMOTE to resolve a **33:1** class imbalance, boosting XGBoost F1 from **78%** to **99%** across **50+** engineered features for reliable binary classification of loan eligibility.
- Delivered a ranked loan product recommendation engine matching **100%** of eligible inquiries to qualifying SBA products at **95%** average confidence, reducing manual lender review time.

Software Engineer, NielsenIQ Feb 2022 – Jun 2024

- Built **4** ingestion pipelines centralizing data across **3** teams into Postgres, Snowflake, Databricks, and Azure Blob Storage, improving cross-functional accessibility and reducing redundant data retrieval.
- Reduced data validation errors by **20%** across **100+** cross-mapped features and cut Spark ETL debug time by **95%** by building Angular and REST API monitoring dashboards with real-time visibility.
- Developed **2** order management services using Java and Spring Boot, optimizing deployments and end-to-end testing by **40%** through Jenkins, SonarQube, and Azure VMSS CI/CD automation.

Data Science Student Assistant, University of Colorado Boulder Apr 2025 – May 2026

- Analyzed internship and job outcome data for **150+** students using Python and Excel, delivering Tableau dashboards that surfaced actionable trends for nontechnical stakeholders in program decision-making.
- Managed communications for **300+** students and led team meetings, improving response time by **30%** and enhancing program engagement and operational efficiency.

Projects

SkillRadar – Multi-Agent GenAI Skill Intelligence System GitHub | Live App | Demo

- Architected a **4-agent** LangGraph DAG integrating live job postings, arXiv papers, and Hacker News signals to forecast skill demand **6 to 12 months** ahead; engineered a Pinecone RAG pipeline (~**450** chunks) with LangSmith observability.
- Designed the Skill Velocity Score (SVS) blending job demand (**50%**), community signals (**30%**), and research trends (**20%**); deployed via FastAPI and Streamlit Cloud with an **80-test** pytest suite.

InsightLens – NLP Review Intelligence Pipeline GitHub | Live App | Demo

- Processed **568K** Amazon reviews through a Sentence-Transformer embedding pipeline achieving **80%** Precision@10 via FAISS semantic search; surfaced **50** KMeans topic clusters with PCA (95% variance retained).
- GPT-4o-mini generates structured insights on themes, strengths, and pain points with TF-IDF as fallback, achieving **90%** actionability and **95%** coverage; evaluated with Precision@10 and nDCG@10 metrics.

Reel Metrics – ML Audience Segmentation System GitHub | Website

- Trained XGBoost and Logistic Regression models achieving **90%** accuracy; applied association rule mining to surface **5+** engagement patterns and dimensionality reduction for segmentation across **3** decades of film data.